

Final Mock Exam — Set A — Solutions

Lectures 1–12 (comprehensive)

Prof. Dr. Christoph Weisser

Quantitative Research Methods · HSBI

Summer Semester 2026

Overview of the final mock exam — Set A

Problem	Topic	Lecture slides	Points
P1	Moving beyond linearity — splines (Ch. 7)	Moving Beyond Linearity (Lec. 9)	16
P2	Generalised additive models (Ch. 7)	Moving Beyond Linearity (Lec. 9)	8
P3	Regression & classification trees (Ch. 8)	Tree-Based Methods (Lec. 10)	20
P4	Ensembles: bagging/RF/boosting (Ch. 8)	Tree-Based Methods (Lec. 10)	16
P5	Neural networks by hand (Ch. 10)	Deep Learning (Lec. 11)	20
P6	Multiple testing (Ch. 13)	Multiple Testing (Lec. 12)	20
P7	Cumulative essentials (Ch. 2–6)	Lectures 1–8	20
Σ			120

Exam conditions

120 minutes for 120 points (≈ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

Problem 1 — task

Problem 1 (16 P) — Moving beyond linearity

- (a) [4 P] Degrees of freedom of a cubic spline with K interior knots — justify by counting parameters and constraints; evaluate for $K = 5$.
- (b) [4 P] Degrees of freedom of a *natural* cubic spline with K knots ($K = 5$); which extra constraint acts where and why is it useful?
- (c) [4 P] Smoothing spline $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$: role of the two terms; behaviour of $\hat{g}$ as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.
- (d) [4 P] Truncated-power basis of a cubic spline with one knot ξ ; define $(x - \xi)_+^3$; why does adding it keep the fit smooth at ξ ?

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Problem 1 — solution (a) degrees of freedom of a cubic spline

Solution P1(a) — count parameters; subtract constraints

Step 1 — unconstrained parameters. K interior knots cut the range of X into $K + 1$ sub-intervals; on each we place a *separate* cubic (4 coefficients):

$$\underbrace{(K + 1)}_{\text{pieces}} \times \underbrace{4}_{\text{coeffs}} = 4K + 4 \quad \text{free parameters.}$$

Step 2 — smoothness constraints. At *each* of the K knots the neighbouring pieces must agree in value, slope and curvature (continuity of g , g' and g''): that is 3 constraints per knot, $3K$ in total.

$$\text{df} = (4K + 4) - 3K = \boxed{K + 4}.$$

Step 3 — evaluate. $K = 5$: $\text{df} = 5 + 4 = 9$.

Cross-check (basis view): the global basis $1, x, x^2, x^3$ (4 functions) plus one truncated power $(x - \xi_k)_+^3$ per knot (K functions) gives $4 + K$ — the same count.

Problem 1 — solution (b) natural cubic spline

Solution P1(b) — the boundary constraint

A *natural* cubic spline adds one requirement: g is forced to be *linear beyond the two boundary knots*, i.e. on the outer regions $X < \xi_1$ and $X > \xi_K$.

Cost of the constraint. “Linear” means the quadratic *and* cubic terms must vanish in each outer region: that removes 2 parameters per region, $2 \times 2 = 4$ in total,

$$\text{df} = (K + 4) - 4 = \boxed{K} \quad \Rightarrow \quad K = 5 : \text{df} = 5.$$

Where it acts & why it helps. It acts only in the two tails — exactly where data are sparse and ordinary polynomials/splines swing wildly. Forcing linearity there *lowers the variance* of predictions at and beyond the boundary, at the price of a little extra bias: a favourable trade near the edges.

Rule of thumb

Cubic spline $\text{df} = K + 4$; natural cubic spline $\text{df} = K$ — the 4 df “saved” are spent on the two linear tails.

Problem 1 — solution (c) smoothing spline

Solution P1(c) — the two terms and the two limits

$$\underbrace{\sum_{i=1}^n (y_i - g(x_i))^2}_{\text{RSS: fidelity to the data}} + \lambda \underbrace{\int g''(t)^2 dt}_{\text{roughness (curvature) penalty}}$$

g'' measures how fast the slope changes (wiggleness); the tuning parameter $\lambda \geq 0$ trades fidelity against smoothness.

- $\lambda \rightarrow 0$: penalty switched off $\Rightarrow \hat{g}$ can be arbitrarily wiggly and *interpolates* every training point (training RSS = 0, severe overfitting; effective df $\rightarrow n$).
- $\lambda \rightarrow \infty$: any curvature becomes infinitely expensive $\Rightarrow g'' \equiv 0$ is enforced $\Rightarrow \hat{g}$ is the *least-squares straight line* (effective df $\rightarrow 2$).

In between, λ (chosen e.g. by LOOCV) steers the effective df smoothly from n down to 2.

Problem 1 — solution (d) truncated-power basis

Solution P1(d) — one knot at ξ

A cubic spline with a single knot ξ is a global cubic plus *one* extra basis function:

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3, \quad (x - \xi)_+^3 = \begin{cases} (x - \xi)^3, & x > \xi \\ 0, & x \leq \xi. \end{cases}$$

Why smoothness survives at ξ . Evaluate the added function and its derivatives at the knot (from the right):

$$(x - \xi)_+^3|_{\xi} = 0, \quad \frac{d}{dx} = 3(x - \xi)_+^2|_{\xi} = 0, \quad \frac{d^2}{dx^2} = 6(x - \xi)_+|_{\xi} = 0,$$

while $\frac{d^3}{dx^3} = 6$ jumps from 0 to 6. So value, slope and curvature match across ξ and only the *third* derivative jumps — precisely the defining smoothness of a cubic spline (continuous g, g', g'').

Problem 2 — task

Problem 2 (8 P) — Generalised additive models

- (a) [3 P] General form of a GAM for quantitative Y ; how does it extend multiple linear regression?
- (b) [3 P] One advantage concerning the interpretation of the f_j ; the main structural limitation — and its repair.
- (c) [2 P] Backfitting in one or two sentences.

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Problem 2 — solution (a) model form

Solution P2(a) — the GAM and how it extends OLS

A generalised additive model for a quantitative response is a sum of one smooth function per predictor:

$$Y = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p) + \varepsilon.$$

How it extends multiple linear regression. OLS is the special case in which every component is forced to be a straight line, $f_j(X_j) = \beta_j X_j$. A GAM *relaxes* this,

$$\text{linear } \beta_j X_j \longrightarrow \text{smooth } f_j(X_j),$$

letting each f_j be any smooth non-linear function (natural spline, smoothing spline, local regression) so it can bend to the data — *while keeping the additive structure*: the predictor effects still simply add up (no product/interaction terms are introduced).

Problem 2 — solution (b) advantage, limitation, repair

Solution P2(b)

Advantage (interpretability). Because the effects are additive, each predictor's contribution is a *single curve* f_j that can be plotted and read one predictor at a time, holding the others fixed — one panel per predictor, as easy to communicate as a row in a coefficient table, yet free to be non-linear.

Limitation. Additivity forbids *interactions*: the shape of the effect of X_j cannot depend on the value of X_k (no predictor can modulate another).

Repair. Add interaction components by hand, for example

- a product term $X_j \times X_k$, or
- a low-dimensional joint smooth $f_{jk}(X_j, X_k)$,

as an extra term in the sum.

Problem 2 — solution (c) backfitting

Solution P2(c) — fitting a GAM one function at a time

Backfitting avoids one huge joint fit by cycling through the functions:

- 1 Initialise $\hat{\beta}_0 = \bar{y}$ and all $\hat{f}_j \equiv 0$.
- 2 For $j = 1, \dots, p$ in turn, form the *partial residuals* that strip out every *other* fitted effect,

$$r_i = y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik}),$$

and re-estimate f_j by smoothing r_i against x_{ij} (spline / local regression), holding all other $\hat{f}_k$ fixed.

- 3 Repeat the sweep until the $\hat{f}_j$ no longer change (convergence).

Every step is just a simple one-dimensional smoothing problem — modular and easy to fit.

Problem 3 — task

Problem 3 (20 P) — Trees by hand

Number of promotional events x and monthly revenue y (1,000 EUR) for $n = 6$ café branches; one split at s : $R_1 = \{x < s\}$ and $R_2 = \{x \geq s\}$ with region-mean predictions.

i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	4	6	8	14	16	18

(a) **[8 P]** Evaluate the cutpoints $s = 2.5$ and $s = 3.5$ by total RSS; which is chosen? (b) **[2 P]** Using the split at $s = 3.5$ from (a): prediction for a new branch with $x = 5$. (c) **[6 P]** Node with 7 “Stay” / 3 “Leave”: proportions; Gini; entropy ($\ln 0.7 = -0.3567$; $\ln 0.3 = -1.2040$); predicted class. (d) **[4 P]** Cost-complexity pruning: role of α in $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$.

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Problem 3 — solution (a) cutpoint $s = 2.5$

Solution P3(a) — part 1 of 2: evaluate $s = 2.5$

Reference (root node). $\bar{y} = \frac{4 + 6 + 8 + 14 + 16 + 18}{6} = \frac{66}{6} = 11$, with root $\text{RSS} = \sum_i (y_i - 11)^2 = 49 + 25 + 9 + 9 + 25 + 49 = 166$.

Split at $s = 2.5$: $R_1 = \{x < 2.5\} = \{1, 2\}$, $R_2 = \{x \geq 2.5\} = \{3, 4, 5, 6\}$. Region means:

$$\hat{y}_{R_1} = \frac{4+6}{2} = 5, \quad \hat{y}_{R_2} = \frac{8+14+16+18}{4} = \frac{56}{4} = 14.$$

Region RSS:

$$\begin{aligned} \text{RSS}_{R_1} &= (4 - 5)^2 + (6 - 5)^2 = 1 + 1 = 2, \\ \text{RSS}_{R_2} &= (8 - 14)^2 + (14 - 14)^2 + (16 - 14)^2 + (18 - 14)^2 = 36 + 0 + 4 + 16 = 56. \\ \Rightarrow \text{RSS}(2.5) &= 2 + 56 = 58. \end{aligned}$$

Problem 3 — solution (a) cutpoint $s = 3.5$ and the choice

Solution P3(a) — part 2 of 2: evaluate $s = 3.5$ and decide

Split at $s = 3.5$: $R_1 = \{1, 2, 3\}$, $R_2 = \{4, 5, 6\}$. Region means:

$$\hat{y}_{R_1} = \frac{4+6+8}{3} = \frac{18}{3} = 6, \quad \hat{y}_{R_2} = \frac{14+16+18}{3} = \frac{48}{3} = 16.$$

Region RSS:

$$\begin{aligned} \text{RSS}_{R_1} &= (4-6)^2 + (6-6)^2 + (8-6)^2 = 4 + 0 + 4 = 8, \\ \text{RSS}_{R_2} &= (14-16)^2 + (16-16)^2 + (18-16)^2 = 4 + 0 + 4 = 8, \\ &\Rightarrow \text{RSS}(3.5) = 8 + 8 = 16. \end{aligned}$$

Decision. Recursive binary splitting is *greedy*: it picks the smallest total RSS. Since $16 < 58$ (both far below the root's 166), it chooses $s = 3.5$, cleanly separating the three low-revenue branches (mean 6) from the three high-revenue ones (mean 16).

Problem 3 — solution (b) prediction for $x = 5$

Solution P3(b)

A regression tree predicts the *mean of the region* into which a new point falls.

With the chosen split at $s = 3.5$: since $x = 5 \geq 3.5$, the new branch lands in $R_2 = \{x \geq 3.5\}$, whose fitted value is $\hat{y}_{R_2} = 16$. Therefore

$$\hat{y}(x=5) = \hat{y}_{R_2} = 16 \quad (\text{i.e. 16,000 EUR per month}).$$

Note every branch with $x \in \{4, 5, 6\}$ gets the *same* prediction 16: a tree produces a piecewise-constant (step) fit, constant within each region.

Problem 3 — solution (c) node impurity: Gini and entropy

Solution P3(c) — node with 7 “Stay” / 3 “Leave”

Class proportions. $\hat{p}_{\text{Stay}} = \frac{7}{10} = 0.7$, $\hat{p}_{\text{Leave}} = \frac{3}{10} = 0.3$.

Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$:

$$G = 0.7(1-0.7) + 0.3(1-0.3) = 0.7 \cdot 0.3 + 0.3 \cdot 0.7 = 0.21 + 0.21 = 0.42.$$

Entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ (use $\ln 0.7 = -0.3567$, $\ln 0.3 = -1.2040$):

$$D = -(0.7(-0.3567) + 0.3(-1.2040)) = 0.2497 + 0.3612 = 0.611.$$

Predicted class. The node predicts the *majority* class, **Stay** (estimated probability 0.7). Both G and D equal 0 for a pure node and are maximal at a 50:50 split — so this node is only *moderately* impure.

Problem 3 — solution (d) cost-complexity pruning

Solution P3(d) — the role of α

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

$\alpha \geq 0$ is the *price charged per terminal node* $|T|$: it balances the subtree's fit (first term) against its size.

- $\alpha = 0$: leaves are free $\Rightarrow$ criterion = training RSS $\Rightarrow$ the full tree T_0 wins (largest, most overfit).
- $\alpha \uparrow$: each leaf must “pay for itself”; a branch whose RSS reduction is worth less than its cost is pruned $\Rightarrow$ a *nested sequence* of ever-smaller subtrees (weakest-link pruning).
- Choosing α : by *cross-validation* — take the α with the smallest CV error, then refit that subtree on all the data. This targets test error, not training fit, balancing bias against variance.

Problem 4 — task

Problem 4 (16 P) — Ensembles

- (a) [6 P] B identically distributed trees with variance σ^2 and pairwise correlation ρ :
 $\text{Var}\left(\frac{1}{B} \sum_b \hat{f}_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. (i) Why does this show variance reduction and what limits it? (ii) Evaluate for $B = 200$ and $\rho = 0.5$ and $\sigma^2 = 9$; compare with one tree. (iii) Limit as $B \rightarrow \infty$?
- (b) [4 P] Random forests: modification at each split; recommended m for *regression* with $p = 15$; why does it improve on bagging (use the formula $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ from (a))?
- (c) [4 P] Boosting via `GradientBoostingRegressor(n_estimators=500, learning_rate=0.01, max_depth=2)`: what does each of the three arguments control, and what is the slow-learning idea (why `learning_rate=0.01`)?
- (d) [2 P] What does the OOB error estimate and why is it free?

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Problem 4 — solution (a)(i) why bagging reduces variance

Solution P4(a)(i) — reading the variance formula

$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)\right) = \underbrace{\rho \sigma^2}_{\text{correlated part}} + \underbrace{\frac{1-\rho}{B} \sigma^2}_{\text{averaged away}}$$

- The **second term** $\frac{1-\rho}{B} \sigma^2 \rightarrow 0$ as B grows: averaging many trees cancels the *independent* part of their variability. This *is* the variance reduction of bagging.
- The **first term** $\rho \sigma^2$ does *not* depend on B : bootstrap trees are grown on overlapping data and share the same strong predictors, so they are positively correlated ($\rho > 0$), and this correlated variance cannot be averaged away.

Hence $\rho \sigma^2$ is the *floor* that limits how far bagging can drive the variance down.

Problem 4 — solution (a)(ii)–(iii) evaluate and take the limit

Solution P4(a) — numbers for $B = 200$; $\rho = 0.5$; $\sigma^2 = 9$

(ii) Evaluate.

$$\text{Var} = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 = 0.5 \cdot 9 + \frac{1-0.5}{200} \cdot 9 = 4.5 + \frac{0.5 \cdot 9}{200} = 4.5 + 0.0225 = 4.5225.$$

Compared with $\sigma^2 = 9$ for a *single* tree, that is a reduction of

$$\frac{9 - 4.5225}{9} = 0.4975 = 49.75\% \quad (\approx 50\%).$$

(iii) **Limit** $B \rightarrow \infty$. The averaged term vanishes and the variance approaches the floor

$$\text{Var} \rightarrow \rho\sigma^2 = 0.5 \cdot 9 = 4.5.$$

Adding more trees never overfits — it only stabilises the average toward this floor.

Problem 4 — solution (b) random forests

Solution P4(b) — decorrelating the trees

Modification. At *each split*, only a fresh *random subset* of m of the p predictors is allowed as split candidates (bagging always considers all p).

Recommended m (regression). $m \approx p/3$, so for $p = 15$:

$$m = \frac{15}{3} = 5 \quad (\text{classification instead uses } m \approx \sqrt{p}).$$

Why it beats bagging. With bagging a dominant predictor tops the first split of nearly every tree, so the trees look alike and ρ is high. Forcing each split to ignore that predictor about $\frac{p-m}{p}$ of the time *decorrelates* the trees, lowering ρ . By part (a) the variance floor is $\rho\sigma^2$, so a smaller ρ lowers exactly the part of the variance that averaging alone cannot remove.

Problem 4 — solution (c) boosting: parameters and slow learning

Solution P4(c) — the three tuning parameters

The three keyword arguments *are* the three tuning parameters of boosting:

- `n_estimators = B` — number of trees added *sequentially*; unlike bagging, boosting *can* overfit if B is too large, so B is chosen by CV.
- `learning_rate =  $\lambda$` — shrinkage (here 0.01): scales each new tree's contribution; smaller $\lambda \Rightarrow$ slower learning $\Rightarrow$ larger B needed.
- `max_depth =  $d$` — splits per tree / interaction depth (here 2): complexity and interaction order; often stumps ($d = 1$) already work well.

Slow-learning idea. Each tree is fitted to the current *residuals*, and only the small fraction λ of it is added, so the model improves gradually and exactly where it still errs, instead of fitting hard in one step — which generalises better. A small `learning_rate = 0.01` implements precisely this (paid for with a larger `n_estimators`).

Problem 4 — solution (d) out-of-bag error

Solution P4(d) — a free test-error estimate

Why $\approx 1/3$ is left out. A bootstrap sample draws n observations *with replacement*, so the chance a given observation is *never* drawn is

$$\left(1 - \frac{1}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-1} \approx 0.368 \approx \frac{1}{3}.$$

So each tree omits about a third of the data — its *out-of-bag* observations.

OOB error. Predict each observation using *only* the trees that did not use it, then aggregate: the resulting error is a valid estimate of the *test error* of the bagged model.

Why it is free. It reuses the bootstrap's by-product — no separate validation set and no cross-validation loop are needed.

Problem 5 — task

Problem 5 (20 P) — Neural networks by hand

Input $x = (2,3)$; hidden ReLU units $g(z) = \max\{0,z\}$; linear output:

$$A_1 = g(1 + 2x_1 - 1x_2) \quad A_2 = g(-2 - 1x_1 + 1x_2) \quad f(x) = -1 + 3A_1 + 2A_2$$

- (a) [8 P] Forward pass step by step: pre-activations; A_1 and A_2 ; output $f(x)$. What did the ReLU do to unit 2 and why are non-linear activations essential?
- (b) [6 P] Logits $z = (2.0, 1.0, -1.0)$; true class 1: all three softmax probabilities and the cross-entropy loss ($e^2 = 7.389$; $e^1 = 2.718$; $e^{-1} = 0.368$; $\ln 10.475 = 2.349$).
- (c) [4 P] Parameter count of a $p = 5 \rightarrow k = 4 \rightarrow K = 3$ network with biases (show both layers).
- (d) [2 P] Conv layer: 28×28 input; $F = 3$; $P = 1$; $S = 1$: output size via $(W - F + 2P)/S + 1$.

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Problem 5 — solution (a) forward pass

Solution P5(a) — push the input through the network

Step 1 — pre-activations (weighted sums).

$$z_1 = 1 + 2x_1 - 1x_2 = 1 + 2 \cdot 2 - 1 \cdot 3 = 2, \quad z_2 = -2 - 1x_1 + 1x_2 = -2 - 1 \cdot 2 + 1 \cdot 3 = -1.$$

Step 2 — ReLU activations $g(z) = \max\{0, z\}$:

$$A_1 = \max\{0, 2\} = 2, \quad A_2 = \max\{0, -1\} = 0.$$

Step 3 — linear output.

$$f(x) = -1 + 3A_1 + 2A_2 = -1 + 3 \cdot 2 + 2 \cdot 0 = -1 + 6 + 0 = 5.$$

Comment. The ReLU *switched unit 2 off* — its negative z_2 is clipped to 0, so it contributes nothing for this input (though it may fire for others). Without such a non-linearity the whole network would collapse into a single linear map of x no matter how deep; the non-linearity is what lets hidden layers build flexible, piecewise-linear shapes.

Problem 5 — solution (b) softmax and cross-entropy

Solution P5(b) — softmax then cross-entropy; true class 1

Step 1 — exponentiate and sum (given $e^2 = 7.389$, $e^1 = 2.718$, $e^{-1} = 0.368$):

$$\sum_k e^{z_k} = 7.389 + 2.718 + 0.368 = 10.475.$$

Step 2 — softmax $\hat{p}_k = e^{z_k} / \sum_j e^{z_j}$:

$$\hat{p}_1 = \frac{7.389}{10.475} = 0.705, \quad \hat{p}_2 = \frac{2.718}{10.475} = 0.259, \quad \hat{p}_3 = \frac{0.368}{10.475} = 0.035,$$

which sum to 1.000 ✓. **Step 3 — cross-entropy** for true class 1 (given $\ln 10.475 = 2.349$):

$$-\ln \hat{p}_1 = -\ln \frac{e^2}{10.475} = -(2 - \ln 10.475) = \ln 10.475 - 2 = 2.349 - 2 = 0.349.$$

Sanity check. A perfect prediction $\hat{p}_1 = 1$ gives loss 0; a uniform guess $\frac{1}{3}$ gives $\ln 3 \approx 1.099$ — here the network is correct and fairly confident, so the loss is small.

Problem 5 — solution (c) parameter count $5 \rightarrow 4 \rightarrow 3$

Solution P5(c) — weights and biases layer by layer

A fully connected layer from a units to b units has $a \cdot b$ weights and b biases (one bias per receiving unit).

Layer	Weights	Biases	Parameters
input $\rightarrow$ hidden ($5 \rightarrow 4$)	$5 \cdot 4 = 20$	4	24
hidden $\rightarrow$ output ($4 \rightarrow 3$)	$4 \cdot 3 = 12$	3	15
total			39

Total = $24 + 15 = 39$ parameters.

(Forgetting the biases would give only $20 + 12 = 32$ — always add one bias per hidden and output unit.)

Problem 5 — solution (d) convolution output size

Solution P5(d) — 28×28 input; $F = 3$; $P = 1$; $S = 1$

Apply the spatial-size formula to the width (the height is identical):

$$\frac{W - F + 2P}{S} + 1 = \frac{28 - 3 + 2 \cdot 1}{1} + 1 = \frac{27}{1} + 1 = 28.$$

So the output feature map is 28×28 — one such map per filter.

Interpretation. Here $-F + 2P = -3 + 2 = -1$ and stride $S = 1$ leave the size unchanged: this is the “same” convolution, with padding chosen so the spatial dimensions are preserved. A stride $S > 1$ would instead shrink the map.

Problem 6 — task

Problem 6 (20 P) — Multiple testing

- (a) **[4 P]** $m = 25$ independent tests at $\alpha = 0.05$; all nulls true: define and compute the FWER $((0.95)^{25} = 0.2774)$; interpret.
- (b) **[7 P]** Six ordered p -values 0.002; 0.007; 0.009; 0.020; 0.040; 0.320 — FWER control at $\alpha = 0.05$ with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?
- (c) **[6 P]** Benjamini–Hochberg at $q = 0.10$ on the same six p -values (0.002; 0.007; 0.009; 0.020; 0.040; 0.320): thresholds jq/m ; rejections; what exactly does FDR control guarantee?
- (d) **[3 P]** Confirmatory phase-III trial with $m = 4$ pre-specified primary endpoints supporting drug approval: FWER or FDR? Justify.

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

Problem 6 — solution (a) FWER for 25 independent tests

Solution P6(a)

Definition. The family-wise error rate is the probability of *at least one* false rejection,

$$\text{FWER} = \Pr(V \geq 1), \quad V = \#\{\text{true nulls that are falsely rejected}\}.$$

Computation. All 25 nulls true, tests independent, each at level 0.05. A single test *avoids* a false rejection with probability $1 - 0.05 = 0.95$, and by independence

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (0.95)^{25} = 1 - 0.2774 = 0.7226.$$

Interpretation. Each test errs only 5% of the time, yet the chance of *some* spurious “discovery” among 25 unadjusted tests is $\approx 72\%$ — multiple testing without adjustment almost guarantees false findings (with $m = 50$: $1 - 0.95^{50} \approx 0.923$).

Problem 6 — solution (b) Bonferroni at $\alpha = 0.05$

Solution P6(b) — part 1 of 2: Bonferroni

Rule. Reject $H_{(j)}$ iff $p_{(j)} \leq \frac{\alpha}{m} = \frac{0.05}{6} = 0.00833$ — one common threshold for all six tests.

Check each ordered p -value against 0.00833:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.007	0.009	0.020	0.040	0.320
$p_{(j)} \leq 0.00833?$	✓	✓	×	×	×	×

0.002 and 0.007 pass; 0.009 > 0.00833 already fails.

$\Rightarrow$ reject $H_{(1)}, H_{(2)}$: **2 rejections.**

Problem 6 — solution (b) Holm step-down, and Holm $\supseteq$ Bonferroni

Solution P6(b) — part 2 of 2: Holm

Rule. Step down $j = 1, 2, \dots$; compare $p_{(j)}$ with $\frac{\alpha}{m-j+1}$ and *stop at the first failure* (the threshold grows as j increases).

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.007	0.009	0.020	0.040	0.320
$\alpha/(m-j+1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
decision	✓	✓	✓	stop	—	—

$0.002 \leq 0.00833$, $0.007 \leq 0.0100$, $0.009 \leq 0.0125$, but $0.020 > 0.0167 \Rightarrow$ stop: reject $H_{(1)}, H_{(2)}, H_{(3)}$: **3 rejections**.

Why Holm $\supseteq$ Bonferroni. Holm's hurdles satisfy $\frac{\alpha}{m-j+1} \geq \frac{\alpha}{m}$ for every j (equal only at $j = 1$), so they are never stricter than Bonferroni's — Holm is uniformly at least as powerful, while still controlling the FWER with no independence assumption.

Problem 6 — solution (c) Benjamini–Hochberg at $q = 0.10$

Solution P6(c) — FDR control

Rule. Compare $p_{(j)}$ with $\frac{jq}{m} = \frac{j \cdot 0.10}{6}$; let L be the *largest* index that passes, and reject $H_{(1)}, \dots, H_{(L)}$.

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.007	0.009	0.020	0.040	0.320
<math(jq m)<="" math=""></math(jq>	0.0167	0.0333	0.0500	0.0667	0.0833	0.1000
$p_{(j)} \leq jq/m?$	yes	yes	yes	yes	yes	no

Largest passing index: $L = 5$ (since $0.040 \leq 0.0833$, while $0.320 > 0.1000$) $\Rightarrow$ reject *all* of $H_{(1)}, \dots, H_{(5)}$: **5 rejections**.

Guarantee. $\text{FDR} = E\left[\frac{V}{\max(R, 1)}\right] \leq q = 0.10$: *on average* at most 10% of the *rejected* hypotheses are false discoveries. It does *not* bound $\Pr(V \geq 1)$ (that would be FWER). Note the nesting: Bonferroni 2 $\subseteq$ Holm 3 $\subseteq$ BH 5 — the procedures grow more liberal.

Problem 6 — solution (d) confirmatory trial: FWER or FDR?

Solution P6(d)

Recommendation: control the **FWER** (e.g. Bonferroni or Holm).

Reasoning.

- This is a *confirmatory* study whose result feeds a *regulatory* decision, so even a *single* false positive is costly — it could lead to approving an ineffective drug.
- FWER control gives the strong guarantee $\Pr(\text{any false rejection}) \leq \alpha$, exactly what regulators require.
- With only $m = 4$ pre-specified endpoints, the power sacrificed by strict FWER control is small.

FDR control would be appropriate for a *large exploratory screen*, where a few false positives are tolerable — not for a small confirmatory endpoint family.

Problem 7 — task

Problem 7 (20 P) — Cumulative essentials

- (a) [4 P] True or false: “with a flexible enough model and enough data the expected *test* MSE at a point can be driven to zero”. Justify.
- (b) [4 P] statsmodels Logit output for passing an exam from hours of study gives coefficient `hours` = 0.8000 ($z = 5.0$, $P < 0.001$; $e^{0.8} = 2.226$): interpret the effect of one more study hour on the *odds*; why is there no single number for the effect on the *probability*?
- (c) [4 P] One reason to prefer 5- or 10-fold CV over LOOCV and one over a single validation split.
- (d) [2 P] $p = 1,200$ SNPs; only about a dozen truly matter: ridge or lasso? Why?
- (e) [6 P] Match to {linear regression; lasso; logistic regression; random forest} — each exactly once: (i) earnings per extra school year with CI; (ii) customer lifetime value from 3,000 features ($n = 250$) with a short driver list; (iii) 30-day readmission yes/no with odds reported to administrators; (iv) bike-share ridership from hundreds of interacting variables; accuracy only.

Lecture slides: Chapters 2–6 (Lectures 1–8).

Problem 7 — solution (a) can test MSE reach zero?

Solution P7(a) — False

The expected test MSE at a point decomposes as

$$\mathbb{E}[(y_0 - \hat{f}(x_0))^2] = \underbrace{\text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2}_{\text{reducible error}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible error}} .$$

Flexibility and more data can drive the *reducible* part (bias and variance) toward 0. But the last term — the *irreducible error* $\text{Var}(\varepsilon) > 0$ — is noise in Y that no model and no sample size can explain.

It is a *hard lower bound*: the expected test MSE can never fall below $\text{Var}(\varepsilon)$. Hence the claim is **false**.

Problem 7 — solution (b) logistic regression: odds vs probability

Solution P7(b) — coefficient on hours = 0.8000

Effect on the odds. With $\ln \frac{p}{1-p} = \beta_0 + \beta_1 \text{ hours}$, one extra hour adds $\hat{\beta}_1 = 0.8$ to the *log-odds*, i.e. *multiplies the odds* of passing by

$$e^{\hat{\beta}_1} = e^{0.8} = 2.226.$$

The odds rise by about +123% (more than double) — the *same* factor at every starting value of hours.

Why no single number for the probability. The map from log-odds to probability is the S-shaped logistic curve. The same +0.8 in log-odds shifts p a lot near $p \approx 0.5$ but barely at all when p is already close to 0 or 1. So the effect on the probability depends on where the candidate sits on the curve — there is no single constant.

Problem 7 — solution (c) why 5- or 10-fold CV

Solution P7(c) — vs. LOOCV and vs. a single split

vs. LOOCV (variance + cost). The n leave-one-out fits use almost identical training sets, so their errors are highly *positively correlated*; averaging strongly correlated quantities barely reduces variance, so the LOOCV estimate has *higher variance* than 5-/10-fold CV — and it costs n fits (vs. k) absent a shortcut formula.

vs. a single validation split (bias + stability). One split depends heavily on the luck of the draw and *overestimates* test error, because only part of the data trains the model (upward bias). k -fold trains on $(k - 1)/k$ of the data and *averages* k estimates $\Rightarrow$ less bias and a more stable number.

$k = 5$ or 10 sits at the sweet spot of this bias-variance trade-off.

Problem 7 — solution (d) ridge or lasso for sparse SNPs

Solution P7(d) — Lasso

With $p = 1,200$ SNPs but only $\approx$ a dozen truly associated, the truth is *sparse*.

- **Lasso** (ℓ_1 penalty) can set coefficients *exactly to zero*, so it performs *variable selection* and returns a small, interpretable model containing (ideally) just the dozen real signals.
- **Ridge** (ℓ_2 penalty) only *shrinks* coefficients toward zero but keeps all 1,200 predictors — no selection, harder to interpret, and effort wasted on the $\sim 1,188$ irrelevant SNPs.

A sparse truth therefore calls for **lasso**.

Problem 7 — solution (e) matching methods to scenarios

Solution P7(e) — each method used exactly once

- (i) **Linear regression** — quantitative response, low dimension; goal is *inference*: an effect size (extra earnings per school year) *with a confidence interval*, exactly what OLS with standard errors provides.
- (ii) **Lasso** — $p = 3,000 \gg n = 250$ rules out OLS; the ℓ_1 penalty regularises and delivers a *short interpretable list* of the main drivers.
- (iii) **Logistic regression** — binary outcome (readmitted yes/no) with interpretable coefficients: $e^{\hat{\beta}_j}$ are *odds ratios* to report to administrators.
- (iv) **Random forest** — hundreds of interacting variables with strong non-linearities; captured automatically for high *accuracy*, and interpretability is not required.